

EMBRYOLOGY

HEART TUBE → CHAMBERS

- Tube week 3 · loops **right** week 4 · first beat week 4 · septation 5–8.
- Septation order: atrial → AV cushions → ventricular → outflow (AP septum, **neural crest**).
- **VSD** = failed membranous IVS closure (most common CHD).
- Secundum ASD = arrested septum secundum. Primum ASD = primum + cushion failure (**Down**).
- **PFO** = failed primum/secundum fusion → cryptogenic stroke; bubble study.
- **TGA** = failed AP *spiraling* · **Truncus** = failed AP *formation* · **TOF** = AP *malalignment* (anterosuperior infundibular shift).
- DiGeorge 22q11.2 → conotruncal + thymic aplasia + hypocalcemia.
- Remnants: ductus venosus → lig. venosum · FO → fossa ovalis · DA → lig. arteriosum.

PHARYNGEAL ARCHES

ARTERIES · NERVES

Arch	Artery	Nerve
1	Maxillary (portion)	CN V
2	Stapedial / hyoid	CN VII
3	Common carotid + prox. ICA	CN IX
4	Aortic arch (L); R subclavian (R)	CN X (sup. laryng.)
6	Prox. pulm. art.; DA (L)	CN X (recurrent laryng.)

FETAL → NEONATAL

SPO₂ & THE FLIP

- Highest SpO₂: **umbilical vein** (~80%) → **IVC** (highest large vessel).
- Birth flip: ↑ **SVR** (placenta off) · ↓ **PVR** (lungs up + ↑O₂) · ↑ LV systemic contribution.
- DA closes via ↑O₂ + ↓PGE₂.
- **PGE1** keeps DA open · **indomethacin** closes it.

CHAMBERS · SURFACE ANATOMY

X-RAY · STAB WOUNDS

- Most anterior: **RV** (stab at left sternal border → tamponade).
- Most posterior: **LA** (enlargement → dysphagia, hoarseness).
- Apex: L 5th ICS MCL — but the lung overlies the LV first.
- PA CXR right border = **RA**; left border = aortic knob → PA → LA appendage → LV.
- Silhouette: RML opacity silhouettes RA; lingula silhouettes LV.
- **LA appendage** = AF thrombus site (pectinate stasis).
- Auscultation: aortic R 2nd ICS · pulmonic L 2nd ICS · tricuspid LLSB · mitral apex.
- IVDU + new murmur + lung cavities → tricuspid IE, *S. aureus*.

CORONARY ARTERIES

TERRITORIES · DOMINANCE · PAPILLARY

- **RCA** → inferior LV, RV, SA (~60%), AV (~90% right-dominant), PDA.
- **LAD** → ant. LV, ant. 2/3 IVS, apex ("widow-maker").
- **LCx** → lateral LV; **PDA in left-dominant** (~5–10%).
- **Dominance rule**: whoever feeds the PDA feeds the AV node.
- ECG → territory: II/III/aVF = RCA · I/aVL/V5–V6 = LCx · V1–V4 = LAD.
- **Posteromedial papillary muscle** = single PDA supply → ruptures 3–5 d post-MI → acute MR + pulmonary edema.
- **Coronary sinus** = posterior AV groove → drains to RA (from L horn of sinus venosus). Dilated CS = persistent L SVC. Biventricular pacing lead path: subclavian → SVC → RA → CS → AV groove → lateral cardiac vein → LV.

CONDUCTION

SA · AV · HIS

- SA: RA, near SVC junction (subepicardial).
- AV: interatrial septum near CS ostium → **triangle of Koch** (tricuspid annulus + tendon of Todaro + CS ostium).
- Speed: *Park At Ventura*, AV → Purkinje > Atria > Ventricles > AV node.
- Inferior MI + Mobitz I → **RCA**.

PERICARDIUM · EFFUSION

TAMPONADE

- "Water-bottle" silhouette + clear lungs.
- ECG: low-voltage QRS + electrical alternans.
- **Beck's triad**: hypotension + JVD + muffled sounds; pulsus paradoxus >10 mmHg.
- Viral cause: coxsackie B, echo.

AORTA — ADULT

STANFORD · ISTHMUS · STEAL

- Arch in order: brachiocephalic → L CC → L subclavian.
- Abdominal: celiac T12 · SMA L1 · renals L1–L2 · gonadal L2 · IMA L3 · iliac bifurcation L4.
- **Stanford A** = ascending → surgery · **Stanford B** = distal to L subclavian → medical.
- **Blunt aortic injury at the isthmus** (lig. arteriosum tethers).
- ICA atheroembolus → ophthalmic → CRA → painless monocular vision loss + cherry-red spot.
- **Subclavian steal**: retrograde ipsilateral vertebral flow + >15 mmHg interarm SBP difference.

ABDOMINAL · PELVIC VEINS

IVC · GONADAL · FEMORAL

- **IVC formed L4–L5** by R+L common iliacs; right of aorta, anterior to vertebra.
- L renal vein crosses anterior to aorta, posterior to SMA → also receives L gonadal & L adrenal.
- R gonadal → IVC · L gonadal → L renal → IVC.
- **Nutcracker** → L flank pain + hematuria ± L varicocele.
- **Right adult varicocele = RCC** invading IVC.
- Femoral triangle (NAVEL lat→med): vein medial to artery.
- Femoral puncture above inguinal lig → retroperitoneal hematoma (inf. epigastric / external iliac).

UPPER-BODY VEINS

SVC · BRACHIOCEPHALIC

- Subclavian + IJ → brachiocephalic; both → SVC → RA.
- **SVC** from common cardinal veins (embryo).
- Unilateral face/arm swelling = brachiocephalic (apical tumor) · bilateral = SVC syndrome.

TOP 15 TRAPS

1. VSD = membranous IVS.
2. PFO + bubble study + cryptogenic stroke.
3. TOF = anterosuperior infundibular shift (neural crest).
4. TGA = failed AP spiraling.
5. PDA = 6th L arch; PGE1 / indomethacin.
6. Coronary sinus = posterior AV groove.
7. Left-dominant → LCx feeds AV node.
8. Posteromedial papillary rupture 3–5 d post-MI.
9. Stanford B near L subclavian.
10. Blunt aortic injury at the isthmus.
11. IVC = L4–L5 from common iliacs.
12. L gonadal → L renal → IVC; nutcracker.
13. Right adult varicocele → RCC + IVC invasion.
14. Subclavian steal: vertebral retrograde.
15. LA appendage = AF clot; fossa ovalis = transseptal puncture site.